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graph TD; VeraCrypt --> USDS[University of San Diego]; VeraCrypt --> CYBR504[CYBR-504 Cryptology]; USDS --> ProfBehar[Professor Behar]; CYBR504 --> Researchers[Julio Arredondo, Jeanelle Jones, & Daniel Mintz];
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VeraCrypt

University of San Diego

CYBR-504 Cryptology

Professor
Behar

Julio
Arredondo,
Jeanelle
Jones, &
Daniel Mintz

OBJECTIVES

- By the end of this lesson, learners will be able to:
 - Explain the importance of data-at-rest encryption
 - Describe VeraCrypt container and full-disk encryption options
 - Create and mount an encrypted VeraCrypt volume
 - Apply secure password and key management practicesIdentify operational security considerations when using encryption

SCENARIO

The Navy-Marine Corps Relief Society (NMCRS) provides financial assistance to service members and their families

Handles sensitive personal and financial information

Staff uses laptops and removable storage (USB drives) across multiple locations

Requires a secure, cost-effective solution to protect data across systems

PROBLEM STATEMENT

Laptops and USB drives can be lost or stolen

Unprotected data can be accessed immediately

Risk of exposing sensitive service member information

Potential impact to privacy, trust, and operations

WHY ENCRYPTION



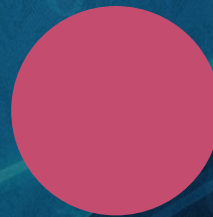
Protects data at rest
from unauthorized
access



Ensures
confidentiality of
sensitive information



Prevents data
exposure if devices
are compromised



Essential for
protecting NMCRS
client data

WHAT IS VERACRYPT

Free, open-source disk encryption software

Supports Windows, macOS, and Linux

Secures data on laptops and removable storage

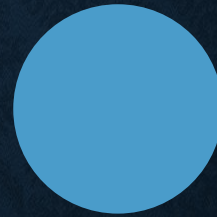
USE CASES

- Securing removable media
- Protecting laptops containing PII
- Secure transport of sensitive files
- Compliance support for sensitive organizational data

TYPES OF ENCRYPTION



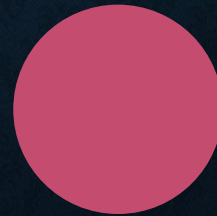
Full-disk protects the entire device, including operating system and all files



Partition secures a specific section of a drive



Container creates an encrypted file that acts as a secure storage space



VeraCrypt supports all three methods depending on the security need

RECOMMENDED APPROACH FOR NMCRS

Container encryption is the recommended solution for NMCRS

Provides secure, portable storage for sensitive data

Works across multiple systems without encrypting the entire device

Well-suited for protecting data on USB drives and external storage

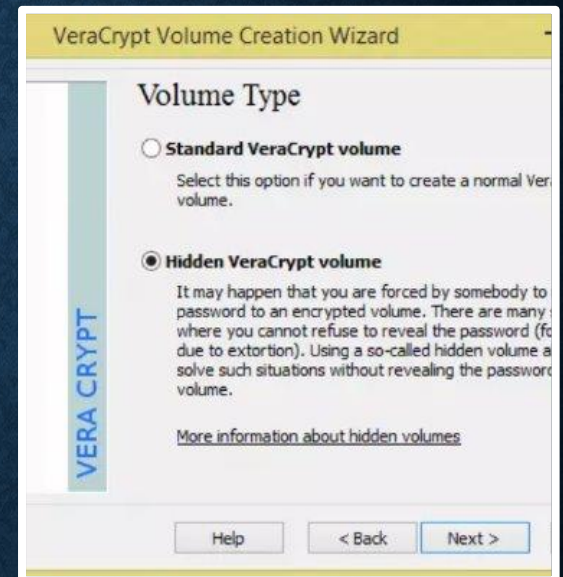
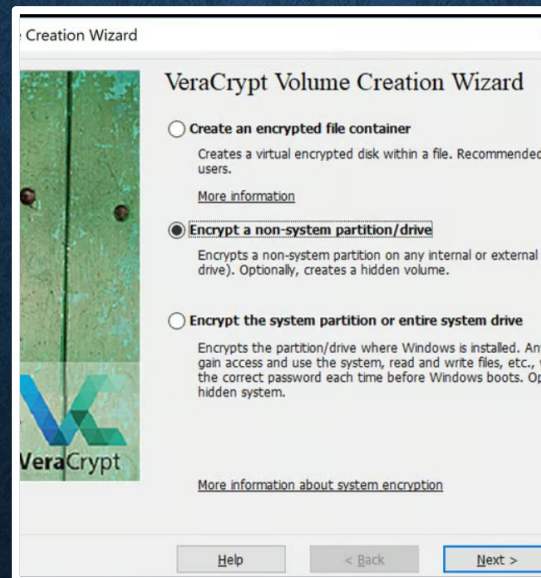
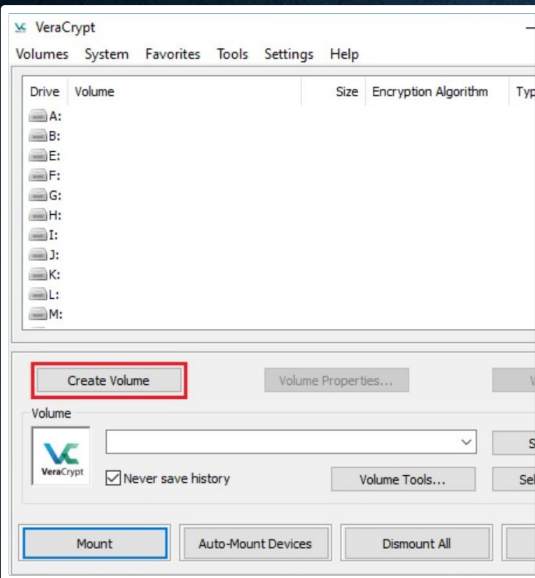
INSTALLATION



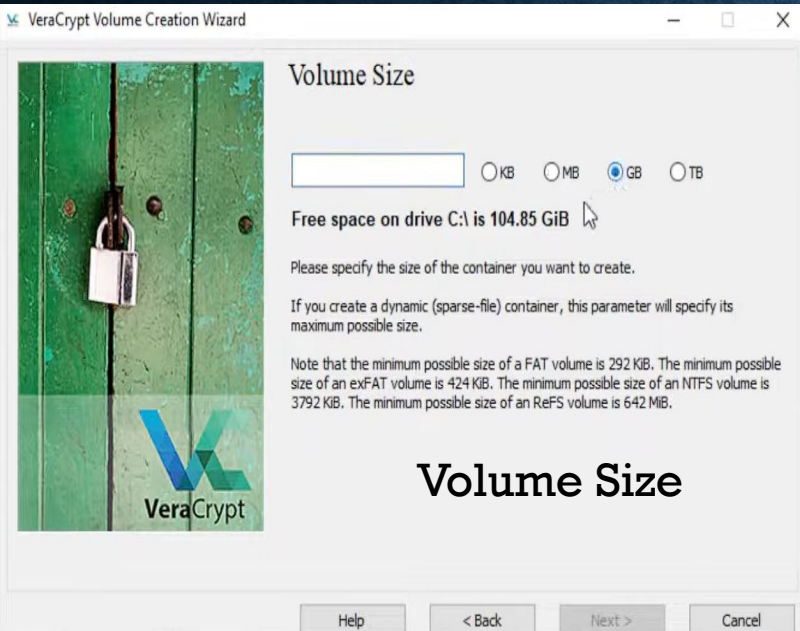
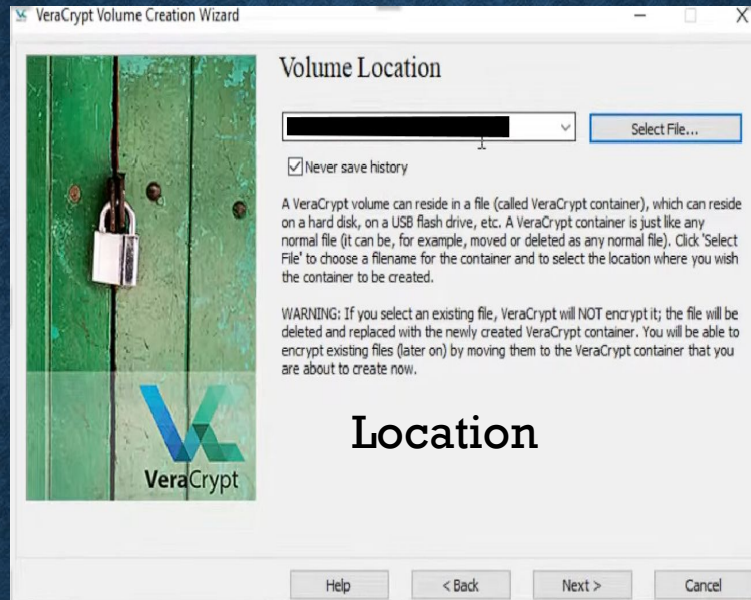
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File Machine View Input Devices Help
Activities Terminal
Apr 4 11:17
julio@julio-VirtualBox: ~
julio@julio-VirtualBox:~$ sudo add-apt-repository ppa:unit193/encryption -y
sudo apt update
sudo apt install veracrypt -y
```

VeraCrypt can be installed using platform specific installers or Linux package managers

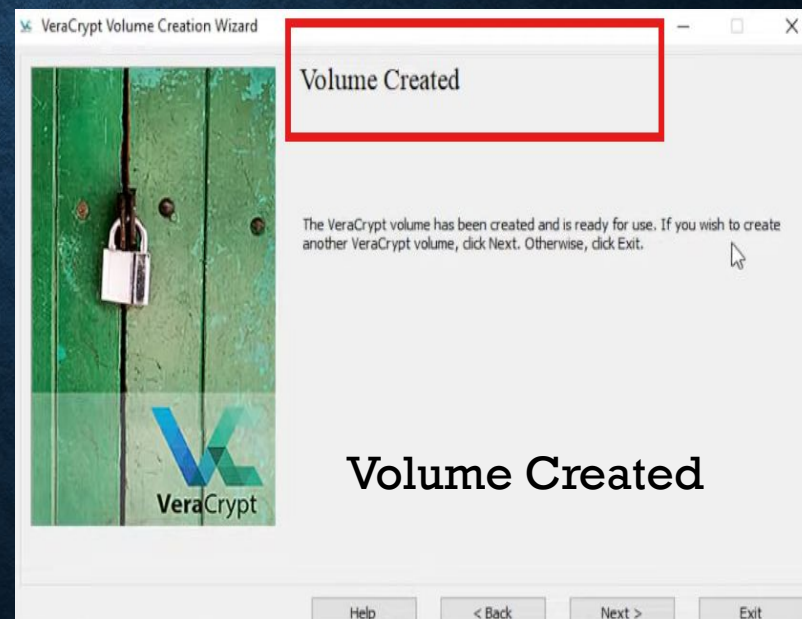
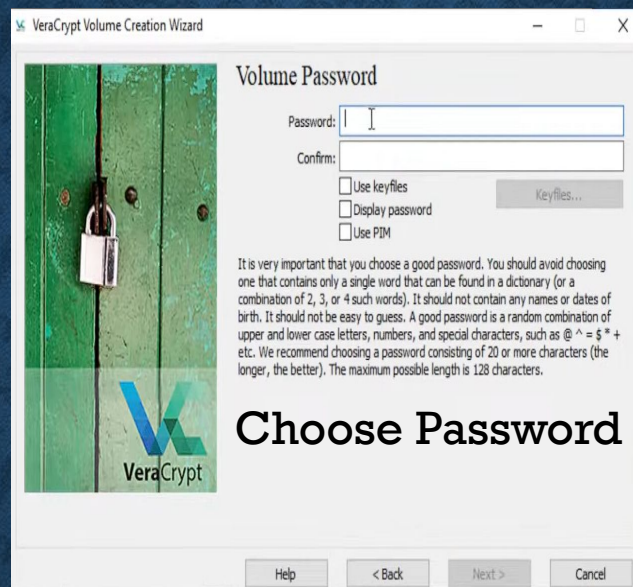
CREATING A VOLUME



CREATING A VOLUME

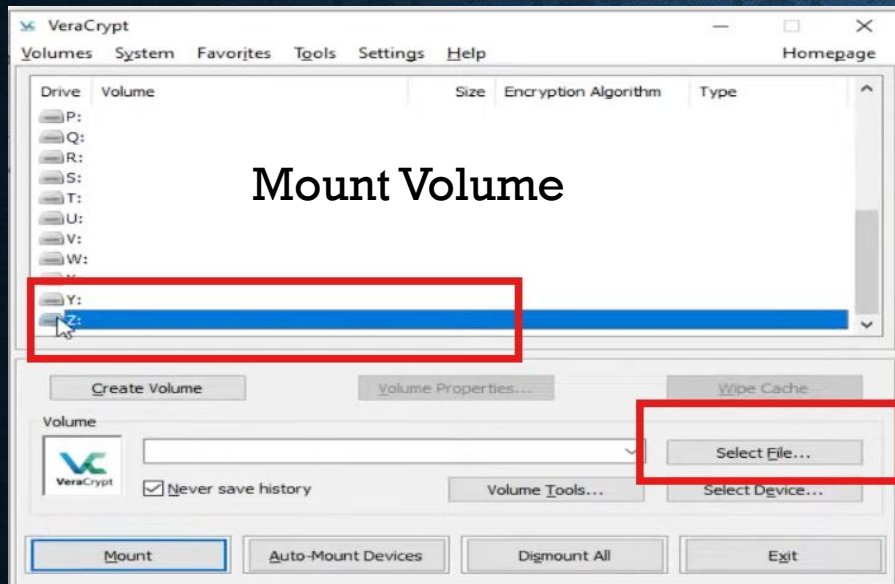


CREATING A VOLUME

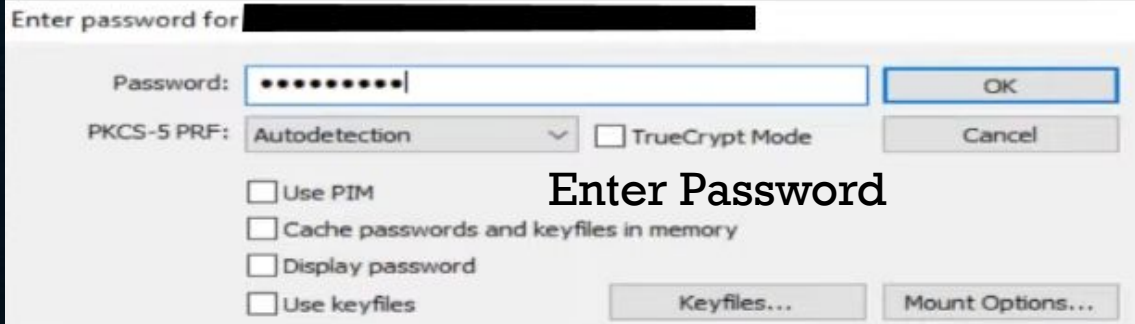
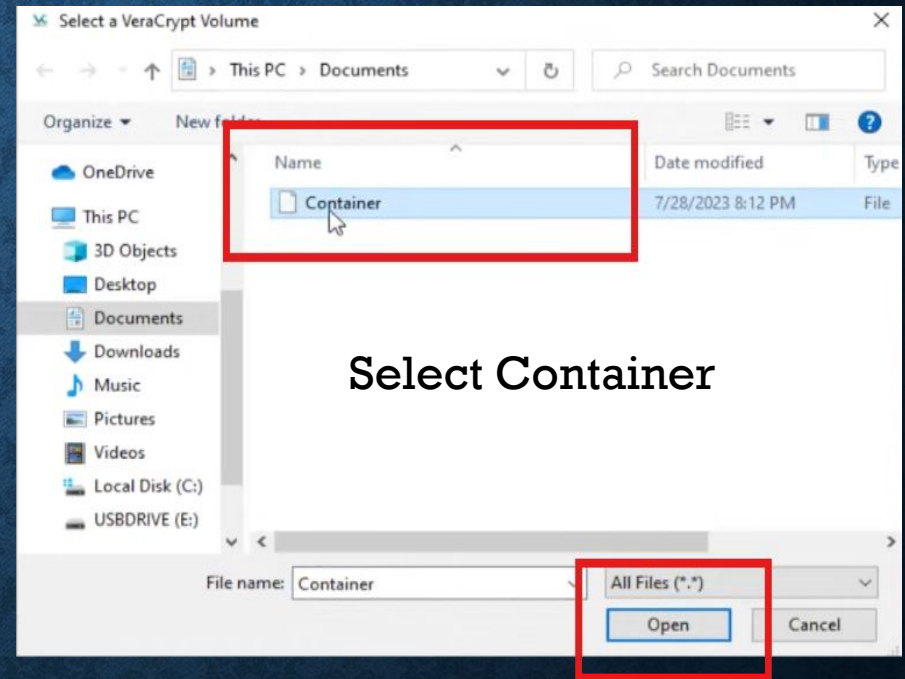


ACCESSING THE ENCRYPTED VOLUME

Mount Volume



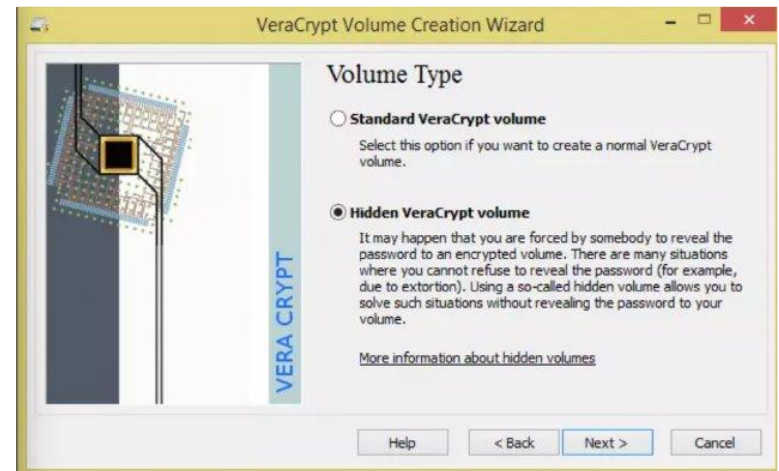
Select Container



Enter Password

HIDDEN VOLUMES

- Hidden volumes add an additional layer of security
- Stored within another encrypted container
- Not visible without the password
- Protects sensitive data from forced disclosure

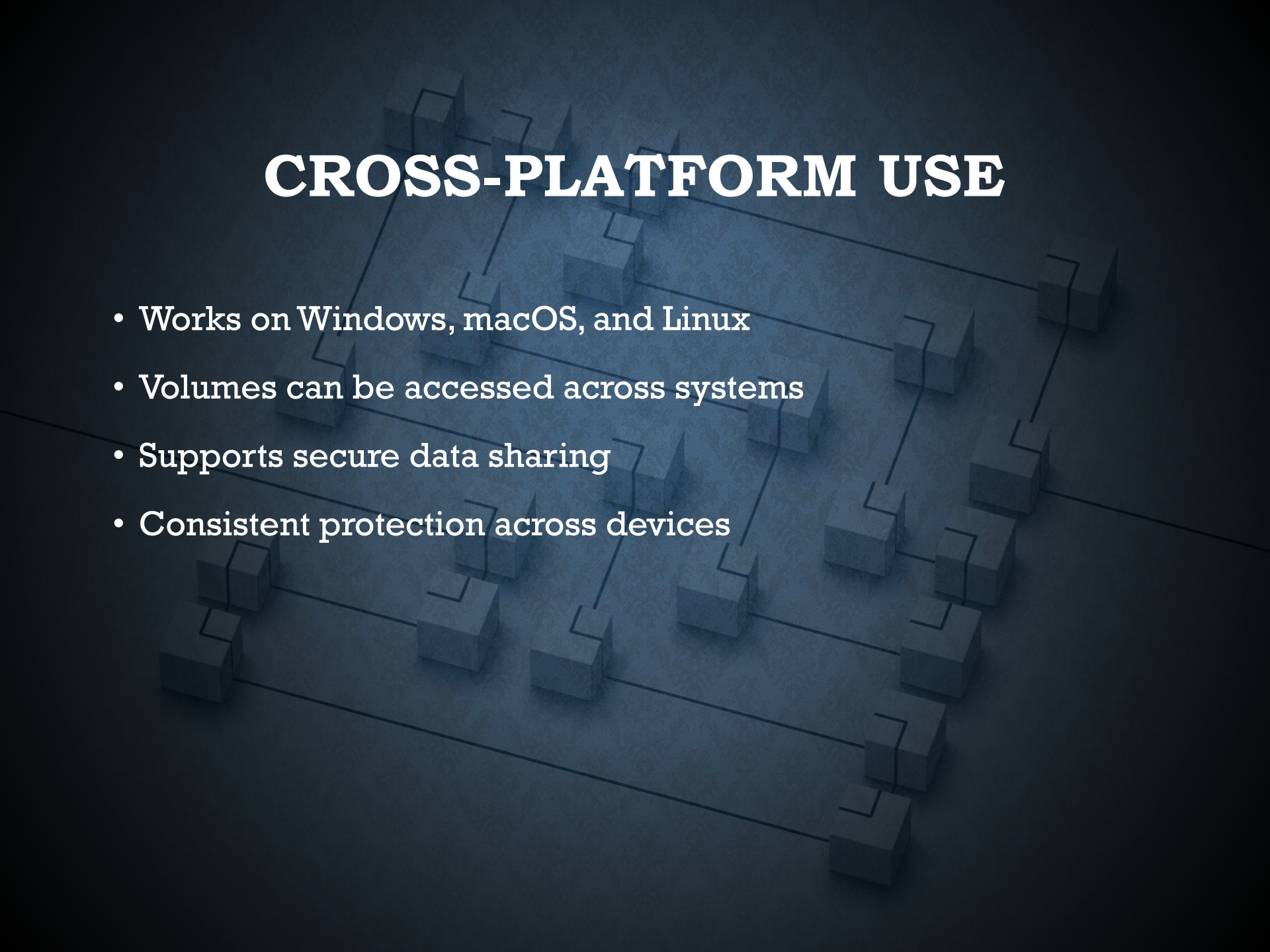




KEY MANAGEMENT

- Use strong, unique passwords
- Do not reuse passwords
- Securely store or back up access credentials
- Loss of credentials results in permanent data loss

CROSS-PLATFORM USE

The background of the slide features a dark blue, almost black, gradient. Overlaid on this are numerous 3D cubes of varying sizes and orientations. These cubes are interconnected by a network of thin, light blue lines, creating a complex, geometric pattern that suggests a digital or networked environment. The lighting on the cubes gives them a three-dimensional appearance, with some faces being more illuminated than others.

- Works on Windows, macOS, and Linux
- Volumes can be accessed across systems
- Supports secure data sharing
- Consistent protection across devices

TROUBLESHOOTING

Lost passwords cannot be recovered



Improper dismounting may cause container corruption

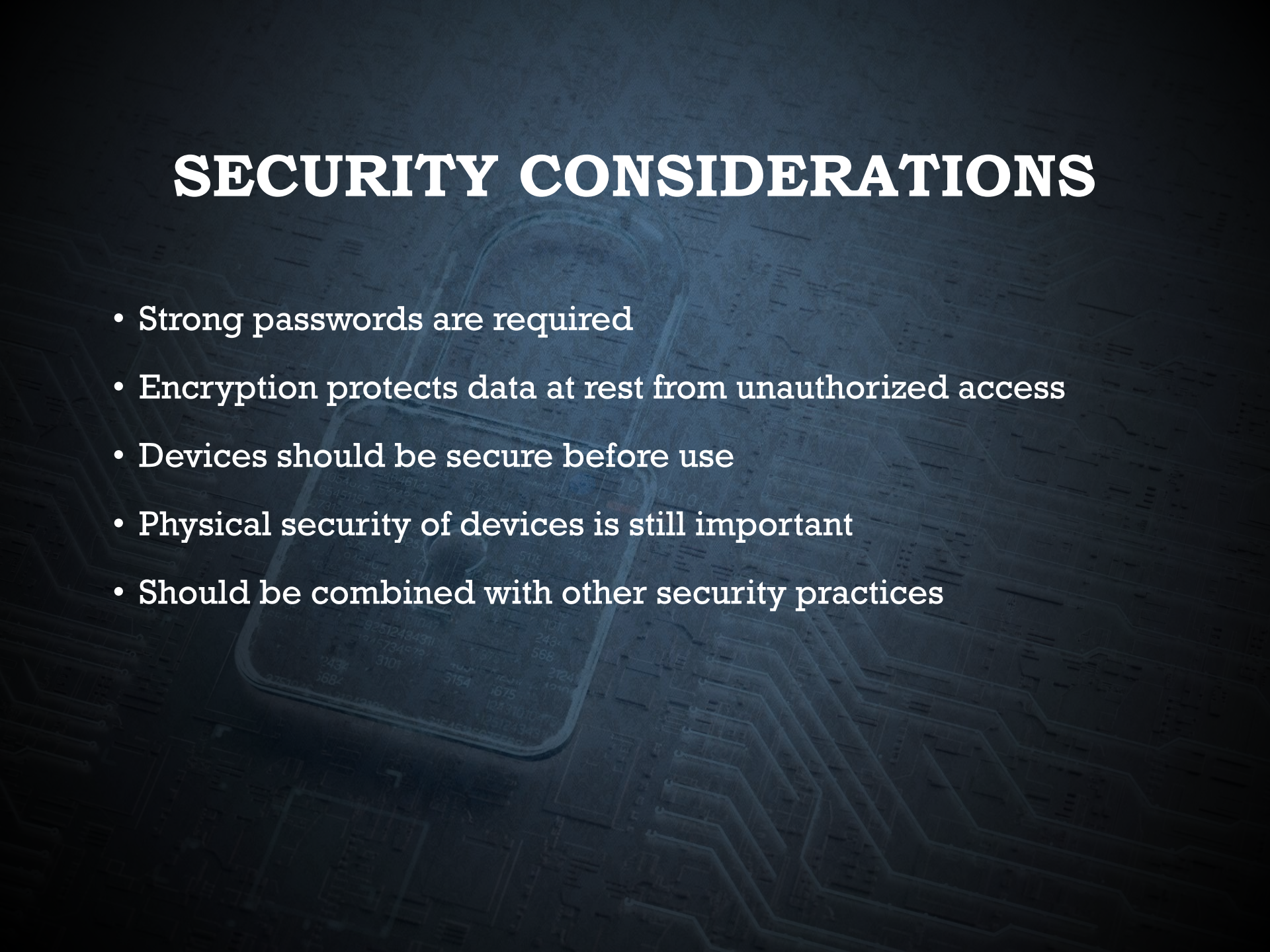


Performance may vary based on system resources



Users must follow proper procedures to maintain data integrity

SECURITY CONSIDERATIONS

The background of the slide is a dark, blue-toned image of a circuit board. A large, rectangular microchip is positioned in the center, slightly below the title. The chip has various markings, including numbers like 9251248431, 921734572, 2438, 3101, 5154, 675, 1231010, and 123124343. The circuit board itself is covered in a complex network of fine lines and larger components, creating a technical and digital aesthetic.

- Strong passwords are required
- Encryption protects data at rest from unauthorized access
- Devices should be secure before use
- Physical security of devices is still important
- Should be combined with other security practices

DEMO

- Encrypted data cannot be accessed directly
- Access is controlled through VeraCrypt
- Data remains secure when not in use

CONCLUSION

- VeraCrypt provides strong protection for laptops and USB drives
- Works across Windows, macOS, and Linux
- Container encryption fits NMCRS' needs
- Protects sensitive data if devices are lost or stolen
- Simple and free security solution



QUESTIONS?